
THE CATALYST



ISSUE 01
JANUARY
2026



ASHFORD SCHOOL
Inspiring Minds

THE CATALYST - CONTENTS

Contents.....	1
Foreword.....	2
Charles Darwin.....	3
Steven Hawking.....	4
Electron Microscopes.....	5
Discovery of Cell.....	6
2025 Physics Nobel Prize.....	7
Under the Modern Scope.....	8
AS Space Experiment.....	9
Ball's Pyramid.....	10



Issue 01 - Sources + More

All the sources used in this edition is visible on our website and there is some more information on **William Lusted**, our guest for this Issue!

<https://sites.google.com/view/thecatalyst-magazine/issue01>

JACOB NAVARATNE

Jacob Navaratne is in Year 9 at Ashford School and he, and William Rees founded The Catalyst.

Jacob joined Ashford School at the start of the Spring Term in Year 8. He enjoys Computer Science, Maths and Physics. He is also an Academic Scholar and really enjoys swimming - like William. He has a brother in the Prep School and is very good friends with both Will and Zach. Jacob and Will both created the idea for THE CATALYST during a Biology project, studying Charles Darwin.

ZACHARY WORDEN

Zachary Worden is an Academic, Music and Drama scholar, he is very passionate aviator, he especially likes the Airbus A350-900ULR, the long range version of the A350-900. He is also an air cadet and an EDI Ambassador He enjoys Maths and Physics, the subjects you need to be a pilot. He is very good at Drama, Music and Maths.

WILLIAM REES

William Rees is also in Year 9 at Ashford School.

Both he, and Jacob Navaratne founded The Catalyst. Will joined in Reception, and has gone through the school. He is Academic Scholar, Drama Scholar and a Sport Scholar. He enjoys Biology and Geography. He swims at Ashford School with Jacob. He is also really good at Biology and enjoys various other subjects. Jacob and Will both created the idea for THE CATALYST during a Biology project, studying Charles Darwin.

WILLIAM LUSTED

William is interested in plants, animals and chemistry. He also attends Ashford School and is an academic scholar. He is great at science and has a great collection of plants in his home.

IN this edition he is talking about Ball's Pyramid.

SPECIAL GUEST!

THREE MAIN FACTS ABOUT CHARLES DARWIN

➔ Brief Biography



Charles Darwin was born on the 12th February 1809. In a House called the Mount located in the Town called Shrewsbury.

Darwin was known famously for his **theory of Evolution by Natural Selection**. Darwin began his thoughts of the theory in the early 1830s, but he only released his theory in **1859** when he was 50 years old. He published this in his seminal work called '**On the Origin of Species**'.

➔ His Death / His Role



Charles Darwin died on the 19th April 1882 in his sleep. It is thought that he died due to a type of heart failure. He also experienced nausea and vomiting on the day that he died.

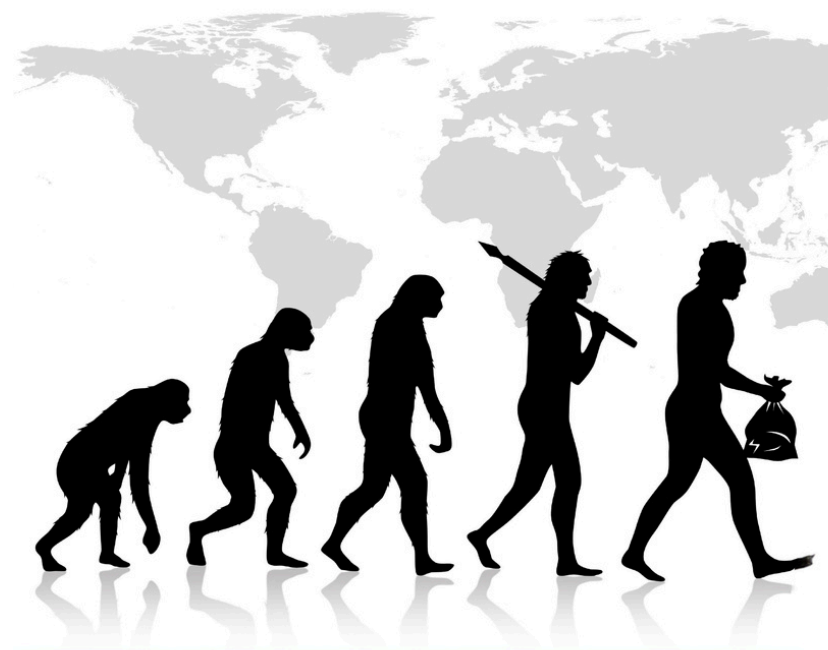
Charles Darwin was not the original naturalist on HMS Beagle. The position was first offered to Leonard Jenyns, who declined. Although unofficial, Darwin served as the ship's naturalist during the voyage. His observations of plants and animals contributed to his theory of evolution by natural selection. The ship's surgeon Robert McCormick was officially picked as the naturalist on board the ship but as the voyage increased Darwin's role as a naturalist increased.

➔ Galapagos Islands



The Galapagos Islands is an archipelago (group of islands grouped together in an ocean) located in the Pacific Ocean. It's located 600 miles west of Ecuador.

They are famous for their wildlife and played a role in Charles Darwin's '**Theory of Evolution**'. The Galapagos Islands are home to unique species found nowhere else in the world such as marine iguanas. The wild animals here are not afraid of humans so it is possible to go closeup with them.



THE NATURALIST WHO SHOOK SCIENCE

By Jacob Navaratne and William Rees - Editors at THE CATALYST

Recently, our investigative reporters at THE CATALYST have received some information about a man named 'Charles Darwin'. He departed from Plymouth for a 5-year trip around South America. Darwin reports to have been nervous and uncertain. Darwin was only a small part of the crew, was only 22 years old when he set off.

Yet this unassuming young man, armed with notebooks and specimen jars, appears to have been tirelessly cataloguing the wonders of the natural world. Reports indicate he has observed a staggering variety of plants, birds, and beasts—some said to defy known classifications. From the dense jungles of Brazil to the stark shores of the Galápagos Islands, Darwin has made countless entries that, sources say, may soon form the basis for a bold new theory.

Whispers from the scientific community hint at a daring concept: that species may change over time through a natural process of selection—where the fittest survive, and others perish. Should such a theory be published, it may stir not only debate, but revolution in our understanding of life itself.

One thing is certain—this young naturalist, once an anonymous cabin guest aboard His Majesty's Ship Beagle, may soon be one of the most talked-about figures in the Empire.



THE MIND THAT REACHED BEYOND THE STARS

By Jacob Navaratne and William Rees - Editors at THE CATALYST

Imagine being able to look into the heart of a black hole or the origins of the universe all from sitting in a wheelchair. Well that was the brilliance of Stephen Hawking, a scientist who defied his limits to explore the deepest mysteries of space and time.

Unlike most cosmologists, Hawking didn't just study the universe—he reshaped how we understand it entirely. His groundbreaking work on black holes revealed that they aren't just space vacuums, but actually emit radiation, now famously known as 'Hawking Radiation'. This discovery challenged old ideas and opened up many new doors in theoretical physics.

But Hawking's genius wasn't bound just to academics. His book 'A Brief History of Time' became a global bestseller, bringing confusing ideas like the Big Bang and quantum mechanics into many living rooms across the world. With a voice synthesized by a computer, he became a global symbol of intellect, resilience, and curiosity. Diagnosed with ALS (Amyotrophic Lateral Sclerosis, also known as motor neurone disease) at just 21, Hawking was told he didn't have long left to live. Instead, he managed to defy that and live on for more than 50 years, continuing to inspire millions of people. His life was a tribute to the power of the human mind—and the refusal to let the biggest limitations define who **YOU** are and could **ever** be.

Thanks to Stephen Hawking, the impossible now seems a little more possible.

A tribute to one of the greatest thinkers the world has ever seen. Rest In Peace

THREE MAIN FACTS ABOUT **STEPHEN HAWKING**

➔ Brief Biography



Stephen Hawking was born on January 8th 1942 in Oxford but moved at an early age to St Albans where he attended St Albans School.

Sadly, Stephen Hawking died 'peacefully' at his house in Cambridge on 14th March 2018 at the age of 76. His funeral took place about a fortnight later on the 31st March at Great St Mary's church in Cambridge. Despite Hawking being an atheist his service was a traditional service.

➔ Computer



As a teenager in the 1950's Stephen built a homemade computer only using discarded parts from clocks telephone switchboards and other salvaged electronics to make a real working computer.

He and his friends even managed to programme it so it could solve basic math equations. This experience with logic help lay the groundwork for his eventual fascination with cosmology.

His classmates recognised his intelligence and called him 'Einstein'. Little did they know it but this had foreshadowed a future of him being one of the most celebrated scientists in recent times.

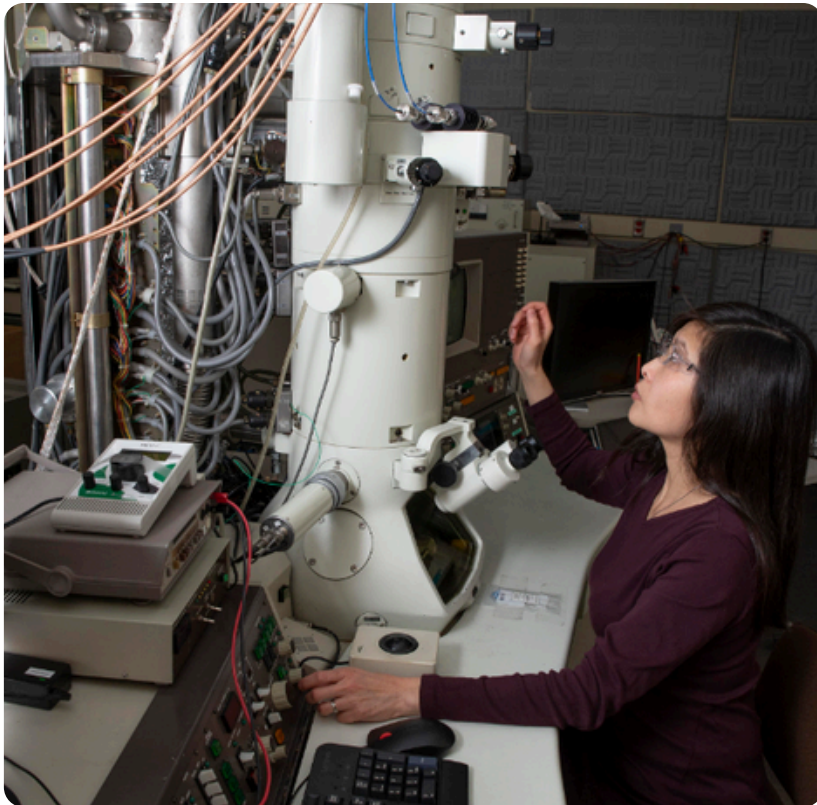
➔ Galileo Luck?



Stephen Hawking's birth on January 8, 1942, marked a symbolic moment in scientific history—it occurred exactly 300 years after the death of Galileo Galilei, the father of modern science. This coincidence links two revolutionary thinkers who, centuries apart, reshaped our humanity's understanding of the universe.

Galileo changed the way people saw the universe entirely. He used a telescope to prove that Earth goes around the Sun, which was a bold idea. Three centuries later, Stephen Hawking also challenged old ideas. He studied black holes and the beginning of the whole universe, helping people understand space in new and unique ways.

Steven Hawking - THE CATALYST



BEYOND THE LIMITS OF LIGHT: EXPLORING WITH ELECTRON MICROSCOPY

By Jacob Navaratne and William Rees - Editors at THE CATALYST

Imagine being able to see some objects hundreds of times smaller than a grain of sand. That is exactly the power of an electron microscope—a revolutionary tool that allows scientists to explore the smallest parts of our world with amazing clarity.

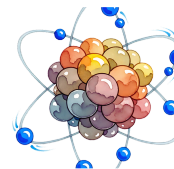
Unlike traditional light microscopes, electron microscopes use beams of electrons to reveal the hidden structures of viruses and atoms. This technology has changed fields like medicine and nanotechnology by uncovering secrets that were once impossible to find with normal light microscopes. Whether it's identifying new diseases or designing materials, electron microscopes open up a gate to a hidden universe.

The masterminds behind this masterpiece were two German inventors, Ernst Ruska and Max Knoll, who, in 1931, built the first electron microscope, changing how we see science forever. Ruska's ground breaking work later earned him the Nobel Prize in Physics, showing his legacy as a pioneer of modern microscopy.

Thanks to their invention, the invisible has now turned visible and science will never ever look the same again.

ELECTRON MICROSCOPES REVEALED...

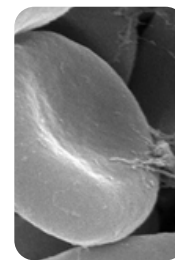
➔ The Two Types



The first type is a **TEM** or **Transmission Electron Microscope**. It works by sending a beam of electrons through ultra -thin specimens to find out about internal structures.

The second type is **SEM** or **Scanning Electron Microscopes**. In SEM's the electrons are sent out and scan the surface of a specimen in order to produce high quality images.

➔ Can you see colour using an Electron Microscope?



While electron microscopes produce high-resolution grayscale images due to electron beams rather than light, any colour seen in published photos is artificially applied during post-processing. This colourization is typically added to improve visual interpretation, but it does not represent the actual colours of the specimen as perceived by the human eye.

This reason is because when electron microscopes blast the electron beam to image a specimen no colour information is recorded.

The area in which the electrons are actually blasted through are white whereas the areas where they weren't are black.

➔ Fascinating Fact

DID YOU KNOW?

Did you know that Ernst Ruska created the electron microscope at the young age of 24? At that point he was just a university student working on his project. However, little did he know that this one invention of his was so powerful that it would shape history forever. Years later he was awarded the Nobel Peace Prize.

Ernst Ruska & Max Knoll - THE CATALYST

THREE MAIN FACTS ABOUT THE **ROBERT HOOKE** AND THE **CELL**

➔ Brief Biography



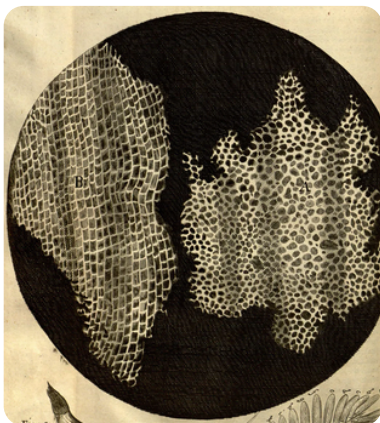
He was born on July 18 July, 1635, Freshwater, Isle of Wight, England—died March 3, 1703.

Hooke had discovered plant cells -- more precisely, what Hooke saw were the cell walls in cork tissue. In fact, it was Hooke who coined the term "cells": the boxlike cells of cork reminded him of the cells of a monastery. Hooke also reported seeing similar structures in wood and in other plants.

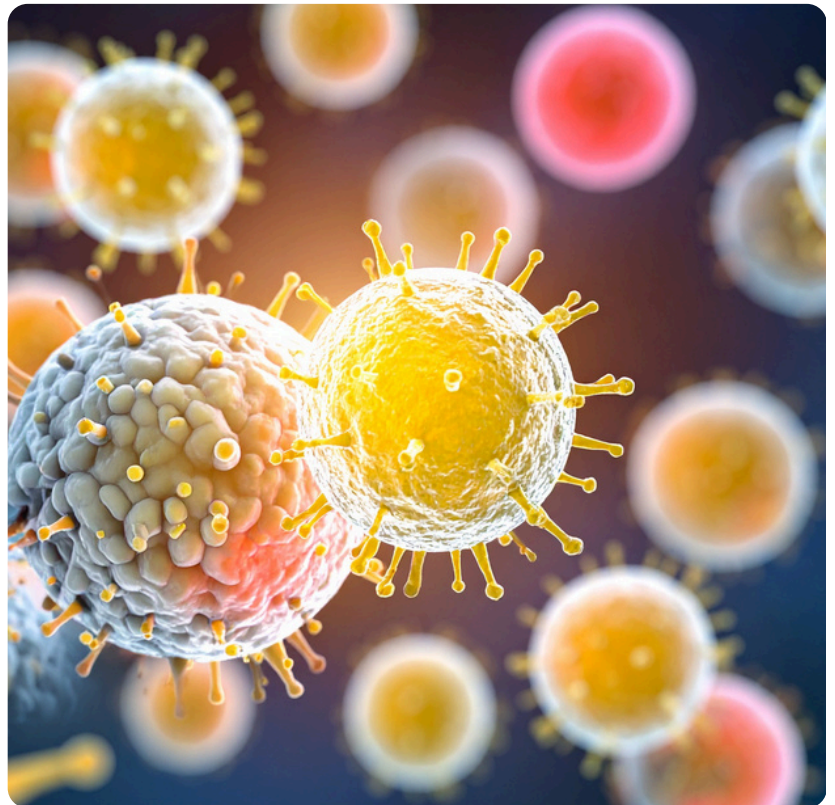
➔ His Microscope:



➔ The First Cells on a Microscope



Robert Hooke - THE CATALYST



THE DISCOVERY OF THE CELL

By Zachary Worden, Journalist at THE CATALYST

Recently, we have investigated the discovery of the cell, and we have revealed lots of facts about the man who found the cell. Robert Hooke. So what did he do?

Firstly, he improved microscope design: Hooke refined the compound microscope, making it powerful enough to reveal structures invisible to the naked eye. Second, he made an observation of a cork, by placing a thin slice of cork under the microscope and noticed it was made up of tiny, box-like compartments. He also coined the term "cell": these compartments reminded him of the small rooms (cells) in a monastery, so he named them "cells".

Furthermore, he published *Micrographia*: In 1665, Hooke recorded his findings in *Micrographia*, a ground breaking book filled with detailed illustrations of microscopic observations. And finally, the foundation for cell theory. Although Hooke only saw dead plant cell walls, his discovery laid the foundation for later scientists to understand that all living things are made of cells.

In 1664, Robert Hooke discovered the massive red spot on Mars. With the information that he discovered, he was able to calculate the length of a Jupiter day.

In conclusion, Robert Hooke has been a crucial part in the development of modern biology. Without him we would be stuck in the old days. His inventions have led us on to discover more about human, plant and animal biology.

THREE FACTS ABOUT THE NOBEL PRIZE 2025

➔ What is the Physics Nobel Prize Award



The Nobel Prize in Physics is an annual award given by the Royal Swedish Academy of Sciences for those who have made the most outstanding contributions to mankind in the field of physics



PHYSICS NOBEL PRIZE 2025

By Zachary Worden - Journalist at THE CATALYST

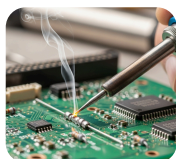
Last year, these names became famous: John Clarke, Michel Devoret and John Martinis. These men have the Physics Nobel Prize 2025. So, what did they win it for?

They won for his incredible discoveries of the macroscopic quantum mechanical tunnelling and energy quantisation in an electric circuit. They demonstrated that the bizarre properties of the quantum world can be made concrete in a system big enough to be held in the hand. Their superconducting electrical system could tunnel from one state to another, as if it were passing straight through a wall. They also showed that the system absorbed and emitted energy in doses of specific sizes, just as predicted by quantum mechanics.

Quantum Physics the research of the smallest things in the universe. This can be confusing because it is very different to normal physics. For example, in normal physics, a ball is either there or not. But in quantum physics, particles can be in many places at a time. These tiny particles are unpredictable and don't follow the normal rules of physics.

They created a computer that can do extremely complicated calculations that a normal computer can't do. They did this by creating SQUIDS (tiny super-sensitive magnetic detectors) and Superconducting circuits.

➔ Superconducting Circuit



A superconducting circuit is a circuit that has 2 components. An inductor (L) and a capacitor (C). these two components bounce back electricity from each

other creating a preferred frequency. Think of it as a tuning fork that vibrates at a certain pitch. This type of circuit dissipates (gives of) no energy. This is because heat can disrupt quantum information. There are a class of artificial atoms that can be used as qubits. Qubits, or quantum bits, are the basic unit of information used to encode data in quantum computing. This is the quantum equivalent of normal bits that normal computers use.

➔ Quantum Computer



Nobel Prize 2025 - THE CATALYST

UNDER THE MODERN SCOPE

SCIENCE >

CURING DISEASES

Know how it sounds! Curing diseases? Well hopefully, I might be possible soon. CRISPR gene editing has reached a new milestone with CRISPR 3.0, a version that's safer, more precise, and capable of editing multiple genes at once.

This leap means scientists can now target and correct the root causes of complex genetic disorders like cystic fibrosis, sickle cell anemia, muscular dystrophy, and even inherited blindness—not just manage symptoms, but potentially cure them.

Unlike earlier versions, CRISPR 3.0 dramatically reduces off-target effects and improves delivery into human cells. Clinical trials in 2025 have already shown success in reversing certain conditions, and researchers believe routine genetic treatments could become part of everyday medicine within the next few years.

This isn't just a cool - but - useless breakthrough—it's a rewrite of what's medically possible.



MEDICINE >

AI MEDICINE DRUG DISCOVERY

AI is revolutionizing drug discovery by using computer models to predict how molecules interact with the body. This speeds up identifying promising drugs for diseases like cancer and Alzheimer's. By analyzing data, AI reveals new treatment targets that traditional methods often miss, transforming medicine development.



AWARDS >

THE NOBEL PRIZE OF 2025

Headlines are full of the Nobel Prize of 2025! These awards range from 'macroscopic quantum mechanical tunneling and energy quantization in electrical circuits (try saying that five times over) and peripheral immune tolerance! The Nobel Prize for Physiology and Medicine this year went to Mary E. Brunkwo, Fred Ramsdell and Shimon Sakaguchi, who 'discovered concerning peripheral immune tolerance'. Their research alone has advanced understand of autoimmune diseases and immune regulation. It doesn't stop there! The Nobel Prize in Chemistry went to Omar M. Yaghi, Susumu Kitagawa and Richard Robson for the development of metal-organic frameworks. These **MOFs** can harvest water from air, even in desert conditions. They act like molecular sponges, trapping water vapour and releasing it again. They mark a significant advancement for saving lives, and is a potential game-changer for water-scarce regions. Read on for more information on the Physics Nobel Prize 2025.



ASHFORD SCHOOL SPACE EXPERIMENT

By Jacob Navaratne - Editor at THE CATALYST

Recently at THECATALYST, we've been looking back at one of Ashford School's most exciting achievements: sending an experiment into space on the Space Shuttle. A team of students designed a small scientific payload that could survive launch, orbit, and landing. They worked with engineers from British Aerospace and NASA to make sure every part was safe, lightweight, and strong enough for space.

Their experiment focused on how certain chemicals react in microgravity. They tested different materials and solutions on Earth first, checking that they were stable inside the stainless steel containers used on the Shuttle. To run everything in orbit, the team even built their own miniature computer using a 65C102 microprocessor, which controlled the experiment and recorded its results.

Communication with NASA took place through computer links and teleconferences, where the team discussed changes, safety requirements, and technical checks. Every modification had to be approved before the experiment could be cleared to fly.

This project showed how students, with determination and expert support, could design a real scientific investigation fit for space. It remains one of the school's most impressive examples of ambition, engineering, and curiosity.

THREE PIECES OF INFO ABOUT THE **AS SPACE EXPERIMENT**

➔ The Experiments



The project aimed to study how certain chemicals (like potassium chromate with copper sulphate, or lead nitrate with potassium iodide)

react and form patterns in space. By doing this in microgravity, they could observe behaviour that can't be seen on Earth.

➔ Built by Students, Approved by NASA



A full science payload for the Space Shuttle was designed, constructed, and tested by Ashford School students. They created everything from the experiment housing to a custom single-board computer that controlled the reactions in orbit. The team worked closely with British Aerospace

engineers and NASA specialists, who checked every component for safety, strength, and reliability. Every revision had to be reported and approved, meaning the students followed the same standards used for real Shuttle hardware.

➔ Student meets Space

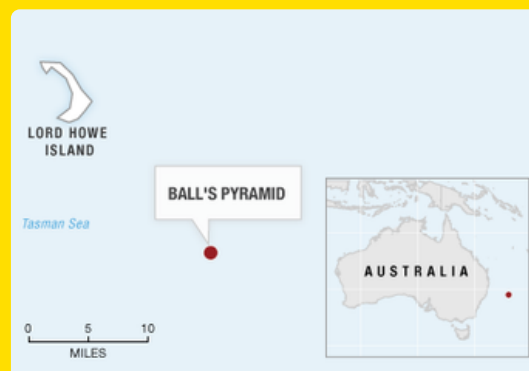


Ashford School - THECATALYST

Ball's Pyramid

Written by William Lusted, friend and guest of THE CATALYST.

Ball's Pyramid is located near Australia and is the tallest volcanic sea stack in the world, standing at 562 metres high and just 300 metres wide. It is largely isolated, and the closest piece of land from it is the rest of the Lord Howe island complex, the closest island being 23km away. It is the eroded basalt remains of a shield volcano formed around 6.4 million years ago. It is known very well though for its incredible preservation of the Lord Howe stick insect or the 'Tree Lobster'. It is one of the largest insects in the world, measuring at around 12 cm long (females). They are dark-brown, nocturnal flightless insects which became extinct on their home of Lord Howe Island after the introduction of black rats from the S.S. Makambo steamship as it passed by the island. They eradicated the population in just 2 years, and the



insects were declared officially extinct in 1920. Miraculously, 81 years later, a team of scientists searched Ball's pyramid and found a tiny group of 24 insects underneath a large shrub. Four were shipped off the island, two destined for a private breeder and two for Melbourne Zoo. After some further research, it was found that Ball's Pyramid grew barely enough vegetation for the insects to live off, but the real mystery was: How did they even get there? Today, it's suspected that they could have rafted there on drifting vegetation, but over 23km, it's incredibly rare. Today, conservation efforts have brought the species up to a population of around 15,000 in captivity, and with any luck, there'll be bright future for these insects and their rare, tiny home.

A Challenge for Only the Bravest and Boldest

Ball's Pyramid is so steep and narrow that only a handful of expert climbers have ever reached the top, and climbing it is now strictly controlled to protect the site. The rock faces drop almost straight into the ocean, which is why it's often compared to a natural skyscraper rising out of the sea. From a distance, it can even look different depending on the light and waves, making it appear taller or thinner than it really is.



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